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PROPOSAL

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1. Introduction

1.1 Research Background

Acute Lymphoblastic Leukemia (ALL) is a cancer affecting the blood and bone marrow, marked by the abnormal proliferation of white blood cells [1]. Among its subtypes, Acute Myeloid Leukemia (AML) stands out due to its complexity and genetic heterogeneity, influencing prognosis and treatment outcomes. Accurate, early diagnosis is essential for personalized therapy [2]. Traditionally, diagnosis relies on microscopic examination of blood or bone marrow smears by haematologists, supported by cytogenetic and molecular tests that detect key genetic abnormalities such as PML-RARA, NPM1, CBFB-MYH11 and RUNX1-RUNX1T1. However, linking visual cellular features to specific genetic subtypes is challenging, requiring expertise and time and is prone to inter-observer variability.

Advancements in artificial intelligence, particularly deep learning (DL), have opened new possibilities for automating Leukemia diagnosis using image-based analysis. Convolutional Neural Networks (CNNs) excel at identifying localized patterns in cytomorphological features. Meanwhile, Vision Transformers (ViTs) provide a broader contextual understanding through self-attention mechanisms [3]. While both architectures show promise, integrating CNNs' spatial sensitivity with ViTs' global perspective remains a critical research direction for improving diagnostic accuracy in complex medical imaging tasks such as AML subtype classification.

1.2 Problem Statement

Despite advancements in machine learning for Leukemia diagnosis, most existing studies are limited to binary classification, differentiating leukemic from normal cells. While valuable, this approach does not meet clinical demands that require precise identification of AML subtypes, each defined by unique genetic alterations and treatment pathways. Subtypes such as PML-RARA, NPM1, CBFB-MYH11 and RUNX1-RUNX1T1 are associated with distinct prognoses and therapeutic responses. Automating their detection from cytomorphological images could significantly improve early diagnosis, reduce reliance on expensive molecular testing and accelerate clinical decision-making.

However, current deep learning models often struggle with multi-class classification and are hindered by class imbalance in medical datasets, where rare subtypes have fewer samples,

increasing the risk of overfitting. CNN-based methods emphasize localized feature extraction but lack global contextual awareness, while ViTs, though powerful in capturing spatial relationships, face limitations in computational cost and fine-grained morphological recognition.

Therefore, an innovative solution is required, one that supports accurate multi-class AML subtype classification, integrates both local and global image features and addresses class imbalance to ensure reliable and generalizable diagnostic performance.

1.3 Research Objectives

This research proposes LEUKEMIACELLNET, a hybrid deep learning model combining CNNs and ViTs for accurate classification of Leukemia blood cells into specific AML subtypes. The core objectives of the study are:

- To develop a hybrid deep learning architecture that integrates CNNs and ViTs for enhanced image-based Leukemia subtype classification.
- To enable multi-class classification of clinically significant AML subtypes, moving beyond traditional binary leukemic vs. healthy detection.
- To implement a Data Expansion and Resampling Strategy (DERS) to address class imbalance and improve model generalization.
- To evaluate the proposed model against existing CNN and transformer-based architectures using metrics such as accuracy, precision, recall and F1-score.

This approach aims to deliver a robust, scalable and cost-effective diagnostic solution that complements or reduces the need for expensive molecular testing, ultimately contributing to improved and personalized Leukemia care.

2. Relevant Works

ALL is the most frequent type of Leukemia in children and it is associated with serious delays and errors in diagnostic processes which is a matter of great concern. Oliveira and Dantas on this problem by using CNNs for classification of leukemic cells [4]. Their research implemented VGG16, VGG19 and Xception architectures and used augmentation techniques

of mirroring, rotation and noise to complement the training and validation datasets. The method rendered an F1-score of 92.60%.

Ansari et al. [5] developed a specialized deep learning architecture for the diagnosis of acute leukaemia based on images of lymphocytes and monocytes. Their model composed of six convolutional layers, four dense layers and a Softmax activation function, was trained on a new dataset improved via GAN. The model wowed many with its accuracy of 99%, it also enabled a faster and more accurate detection than other models, showcasing its ability to be of assistance to healthcare providers in their work.

Karim et al. [6] presented an ensemble machine learning model, called LDSVM, which has been developed to classify leukaemia using DNA microarrays via logistic regression, decision trees and support vector machines. Methods such as k-fold cross-validation, grid search optimization, SMOTE and PCA were used in the study to increase prediction accuracy, tackle class imbalance as well as feature reduction problems. The accuracy, precision as well as the time used to run the proposed LDSVM model was better than the defaults.

Alsaykhan et al. [7] proposed a hybrid diagnosis model that integrates support vector machine (SVM) and particle swarm optimization (PSO) for ALL diagnosis. By using publicly available datasets, the proposed technique enhanced the classification SVM performance that was improved using PSO optimization. The hybrid model showed better detection rates as well as confusion matrix results and therefore, showcased efficiency in improving automated diagnosis devices' aspects relating to medical practices.

Fauzi et al. [8] have investigated the application of Fuzzy Support Vector Machines (FSVM) and its combination with Principal Component Analysis (PCA) for classification of Leukemia. The research was able to obtain an accuracy of 96.92% with PCA as compared to 87.69% obtained in the absence of PCA, hence showing the relevance of PCA in improving classification performance of a model by decreasing non relevant features and targeting crucial ones.

Saikia et al. [9] designed a Whale Optimization Algorithm-based Support Vector Machine (WOA-SVM) for the detection of ALL from microscopic blood smear images. Their method combined 2D-Discrete Orthonormal S-Transform and Local Binary Pattern analysis and segmentation methods such as K-mean clustering. The study also provided evidence showing

the effectiveness of WOA-SVM over traditional methods by achieving 98.42 percent in the ALL-IDB1 dataset.

Table I: Comparison of Relevant Literature

Authors	Dataset	Model	Results	Findings
Oliveira and Dantas	C-NMC 2019 Dataset	CNNs (VGG16, VGG19, Xception)	F1-score: 92.60%	Strong augmentation; limited to F1-score evaluation, binary classification
Ansari et al.	GAN-augmented lymphocyte & monocyte image on self-collected dataset	6 CNN layers, 4 Dense layers, Softmax	Accuracy: 99.5%	High accuracy, fast, limited to binary classification
Karim et al.	DNA microarrays	LDSVM (Logistic Regression + DT + SVM)	Accuracy: 99.5%	Good optimization and imbalance handling; less focus on image data, binary classification
Alsaykhan et al.	ALL-IDB1 and ALL-IDB2	SVM + Particle Swarm Optimization (PSO)	Accuracy: 0.9242%, 0.974%	Efficient hybrid model; lacks deep learning integration, binary classification
Fauzi et al.	Leukemia dataset from gems-system.org	FSVM + PCA	Accuracy with PCA: 96.92%, without PCA: 87.69%	PCA boosts performance; classical model limitations, binary classification
Saikia et al.	ALL-IDB1	WOA-SVM	Accuracy: 98.42%	Effective feature fusion and optimization; traditional ML model, binary classification

While numerous studies have applied machine learning and deep learning techniques to Leukemia diagnosis and achieved excellent performance, the vast majority have concentrated on binary classification tasks, typically distinguishing leukemic cells from healthy ones or differentiating between ALL and AML. Although such binary classification models demonstrate promising accuracy, they fail to address the critical clinical need for comprehensive subtype-level diagnosis within AML, where distinct genetic and molecular subtypes dictate varied prognoses and treatment regimens.

Current state-of-the-art approaches predominantly overlook the granularity required for precise AML subtype classification, which is pivotal for personalized medicine. The subtypes, characterized by molecular markers such as PML-RARA, NPM1 mutations, CBFB-MYH11 and RUNX1-RUNX1T1, exhibit subtle yet diagnostically significant cytomorphological differences. Automated systems focusing solely on binary differentiation do not capture these nuanced features, limiting their applicability in clinical workflows that increasingly demand detailed subtype identification for therapy tailoring.

Furthermore, medical image datasets suffer significantly from class imbalance, where rare AML subtypes are underrepresented, posing challenges to the development of generalized and robust models. Most existing deep learning models do not sufficiently address this imbalance, resulting in overfitting toward majority classes and poor predictive performance on minority subtypes.

The present research addresses these gaps by proposing a hybrid CNN-ViT architecture supported by a DERS that tackles class imbalance and enhances feature learning across scales. This enables accurate multi-class classification of AML subtypes directly from cytomorphological images, offering a meaningful advancement toward automated, precise and clinically relevant Leukemia diagnosis beyond binary classification.

3. Proposed Methodology

3.1 Dataset Description

The foundation of accurate Leukemia subtype classification lies in the availability of a comprehensive, well-annotated image dataset. This study utilizes the AML-Cytomorphology_LMU dataset, which contains high-quality microscopic images of bone marrow and peripheral blood smears [10]. The dataset is specifically annotated with five clinically significant subtypes relevant for AML diagnosis:

- **Normal:** Healthy, non-leukemic bone marrow cells serving as the control group.
- **PML-RARA:** Characteristic of Acute Promyelocytic Leukemia (APL), associated with the t(15;17) translocation.
- **NPM1:** AML subtype bearing Nucleophosmin 1 mutations, prevalent and prognostically important.

- **CBFB-MYH11:** A core-binding factor AML subtype caused by chromosomal inversion $inv(16)$, generally linked to favourable prognosis.
- **RUNX1-RUNX1T1:** Another core-binding factor AML subtype resulting from $t(8;21)$ translocation.

This dataset encapsulates subtle cytomorphological variations linked to each subtype, such as changes in cell shape, granulation and nuclear characteristics, which are critical for precise classification. However, it is inherently imbalanced due to the rarer occurrence of some subtypes, necessitating thoughtful data handling strategies to avoid model bias.

Prior to training, all images undergo rigorous pre-processing, including normalization through resizing and rescaling to consistent dimensions using a fixed `IMAGE_SIZE` parameter. This standardization is crucial to ensure uniform input for convolution-based models and reduce computational overhead. Data augmentation techniques such as rotation, flipping and color jittering are also applied to enhance model robustness, especially for underrepresented classes.

3.2 Model Description

To effectively classify AML subtypes from cytomorphological images, this study introduces LEUKEMIACELLNET, a hybrid architecture merging the strengths of CNNs and ViTs.

- The CNN backbone, based on ResNet50, is responsible for extracting detailed local features such as granular internal structures, cellular shape and texture. Residual connections in ResNet facilitate gradient flow in deep networks, enabling learning of complex morphological patterns essential for subtype discrimination.
- The ViT module complements CNN outputs by modelling global dependencies and spatial relationships within images. Self-attention mechanisms in ViT allow the model to understand how cellular components relate across the micro and macro scales, capturing positional information beyond local neighbourhoods.

One critical innovation is the incorporation of a Data Expansion and Resampling Strategy (DERS) designed to alleviate class imbalance. This method synthetically generates additional samples for majority classes during training, effectively upscaling imbalanced datasets and allowing the model to generalize better across all subtypes. By providing a more

representative and varied training set, DERS helps prevent overfitting on common classes and improves sensitivity for rare categories.

During training, model parameters are optimized through iterative backpropagation using batches containing balanced representation from all classes post-DERS adjustment. Standard performance metrics including accuracy, recall, precision and F1-score are employed for evaluation, ensuring comprehensive assessment of the model's diagnostic capabilities. The overall workflow diagram of this study has been shown in Figure 1.

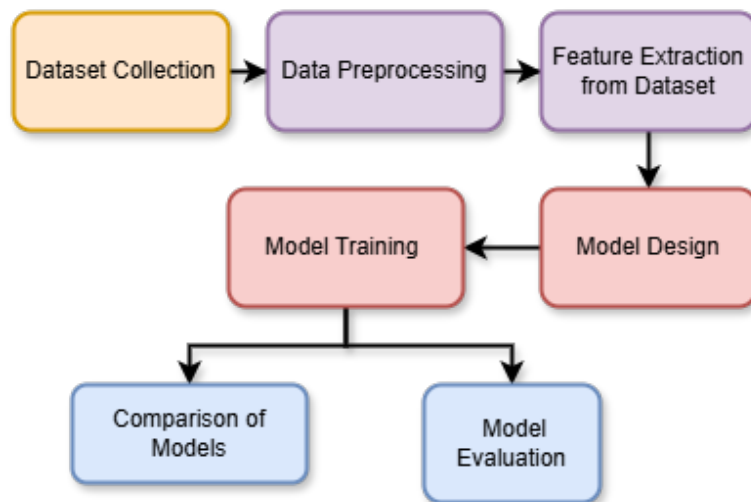


Figure 1: Workflow diagram of the proposed study

This hybrid approach aims to surpass limitations of standalone CNN or transformer models by synthesizing local detail extraction with a global contextual understanding, thereby providing a powerful framework for accurate and clinically relevant multi-class AML subtype classification.

4. Discussion and Conclusion

This research proposes a novel hybrid deep learning framework that advances Leukemia diagnosis by focusing on multi-subclass classification of Acute Myeloid Leukemia (AML), addressing a critical clinical need underserved by existing models. Whereas most prior studies emphasize binary classification to differentiate leukemic from healthy cells, this work targets the nuanced identification of multiple AML subtypes, Normal, PML-RARA, NPM1, CBFB-MYH11 and RUNX1-RUNX1T1, each with distinct prognostic and therapeutic significance.

The proposed hybrid architecture synergistically combines CNNs for detailed capture of cellular morphology with ViTs for modelling global spatial contexts, enabling the recognition of subtle cytomorphological differences across subtypes. Additionally, the integration of a Data Expansion and Resampling Strategy (DERS) directly addresses the prevalent class imbalance in medical imaging datasets, enhancing model generalizability and robustness, especially for rare subtypes.

This innovative approach is expected to significantly improve classification accuracy and clinical relevance compared to existing binary or single-architecture models. By enabling reliable, automated subtype-specific Leukemia diagnosis from cytological images, the study lays the groundwork for AI-driven personalized treatment strategies and improved patient outcomes. Its successful implementation would mark an important contribution toward scalable and precise Leukemia diagnostic tools in clinical practice.

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